Chapter 12: Neuroendocrine and Neuroimmune Systems — Mathematical Equations

Comprehensive Neuroscience Reference

Abstract

This chapter presents the quantitative framework for understanding neuroendocrine axes, immune signaling, and circadian regulation of brain function. We derive kinetic models for blood-brain barrier transport (carrier-mediated Michaelis-Menten kinetics, lipophilicity-dependent diffusion), HPA axis dynamics (cortisol negative feedback, diurnal oscillations, dexamethasone suppression), HPG axis pulsatility and estradiol-driven LH surges, HPT axis suppression, microglial cytokine production and neuronal excitotoxicity, short-chain fatty acid production from fiber fermentation, circadian clock gene regulation (CLOCK/BMAL1 transcription-translation feedback), and melatonin synthesis and degradation. Six case studies with numerical solutions demonstrate: BBB glucose saturation kinetics in hypoglycemia, HPA axis response to acute stress and recovery, LH surge timing during menstrual cycle, TSH suppression in hyperthyroidism, microglial IL-1β production in neuroinflammation, and menstrual cycle effects on hippocampal spine density. All equations include typical parameter values and mechanistic explanations grounded in neurobiology and endocrinology.

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1. Blood-Brain Barrier Transport Kinetics

1.1 Carrier-Mediated Transport (Michaelis-Menten)

The rate of transport across the BBB for substrates requiring specific carriers (e.g., glucose via GLUT1, amino acids via LAT1):

$$v = \frac{V_{\max}[S]}{K_m + [S]} \tag{1}$$

Symbol	Meaning	Typical Value
v	Transport velocity (flux)	50–200 nmol·cm^{-2}·min^{-1}
$V_{\max}$	Maximum transport rate	Glucose: 50–200 nmol·cm^{-2}·min^{-1}
$[S]$	Substrate concentration	5 mmol/L (glucose in blood)
K_m	Michaelis constant (half-saturation)	1.7 mmol/L (glucose/GLUT1)

Explanation: As substrate concentration increases, transport accelerates initially (first-order) but plateaus as carriers become saturated. At plasma glucose (∼5 mmol/L), GLUT1 operates near saturation, ensuring reliable glucose supply to brain despite fluctuations in blood levels.

1.2 Paracellular Diffusion and Lipophilicity

For passive transcellular diffusion through endothelial lipid bilayers:

$$P = P_0 \cdot \log(K_{ow})^{-0.5} \quad \text{or simplified} \quad P = k_p \cdot e^{-0.001 \cdot MW} \tag{2}$$

Symbol	Meaning	Typical Value
P	Permeability coefficient	10^{-7} (hydrophilic) to 10^{-5} (lipophilic)

Symbol	Meaning	Typical Value
K_{ow}	Octanol-water partition coefficient	0.01 (glucose) to 400 (lipophilic drugs)
MW	Molecular weight	Da
k_p	Empirical permeability coefficient	∼0.001 cm/s

Explanation: Lipophilic molecules (ethanol, anesthetics, lipid-soluble drugs) cross the BBB readily via transcellular diffusion; hydrophilic molecules are excluded. Molecular weight >400–500 Da without specific transporters cannot cross.

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2. HPA Axis Dynamics

2.1 Cortisol-Mediated Negative Feedback

The suppression of CRH/ACTH release by circulating cortisol follows a sigmoidal (Hill) dose-response relationship:

$$\text{CRH release} = \text{CRH}_{\text{basal}} \cdot \left(1 - \frac{[Cortisol]^n}{IC_{50}^n + [Cortisol]^n}\right) \tag{3}$$

Symbol	Meaning	Typical Value
$\text{CRH}_{\text{basal}}$	Basal CRH secretion rate	1.0 (arbitrary units)
$[Cortisol]$	Plasma cortisol concentration	0.2–0.5 (basal); 0.5–1.0 (stress)
IC_{50}	Cortisol concentration suppressing CRH by 50%	∼0.15–0.25 μmol/L
n	Hill coefficient (steepness)	1.5–2.0

Explanation: Cortisol inhibits both CRH (via parvocellular neurons) and ACTH (via corticotrophs) in a dose-dependent manner. At basal cortisol (∼0.2 μmol/L), ∼20% suppression occurs. At stress levels (∼0.8 μmol/L), ∼90% suppression is reached. The Hill coefficient

>1 indicates cooperativity — steep suppression within a narrow cortisol range, allowing sharp on/off control.

2.2 Diurnal Cortisol Rhythm

Cortisol exhibits a characteristic circadian oscillation with highest levels at dawn and lowest at midnight:

$$[Cortisol](t) = A_0 e^{-\lambda t} + A_c \cos\left(\frac{2\pi t}{T}\right) + C_{\min} \tag{4}$$

Symbol	**Meaning**	**Typical Value**
A_0	Amplitude of exponential decline	∼0.2 μmol/L
λ	Exponential decay rate	∼0.03 h^{-1}
A_c	Amplitude of circadian oscillation	∼0.15 μmol/L
T	Period (24 h)	86,400 s
$C_{\min}$	Minimum cortisol level (midnight)	∼0.05 μmol/L

Explanation: The diurnal pattern emerges from SCN-driven circadian control of ACTH release via CRH. At ∼6–8 AM, cortisol peaks (∼0.5 μmol/L; facilitating awakening, metabolism, stress preparedness). By 11 PM, levels fall to ∼0.05 μmol/L, promoting sleep.

2.3 Dexamethasone Suppression Test (DST)

$$\frac{[Cortisol]_{\text{suppressed}}}{[Cortisol]_{\text{baseline}}} = \frac{1}{1 + ([Dex]/IC_{50,Dex})^{n_{Dex}}} \tag{5}$$

Symbol	**Meaning**	**Typical Value**
$[Dex]$	Dexamethasone dose	0.5–8 μg/kg
$IC_{50,Dex}$	Dexamethasone IC_{50} for GR	∼0.001 μmol/L
$[Cortisol]_{\text{baseline}}$	Pre-dexamethasone cortisol (8 AM)	0.4–0.5 μmol/L
$[Cortisol]_{\text{suppressed}}$	Cortisol post-dexamethasone	Expected <0.1 μmol/L

Explanation: In healthy individuals, 0.5 μg/kg dexamethasone suppresses next-morning cortisol to <0.1 μmol/L. Non-suppression indicates DST resistance, seen in major depression, Cushing's syndrome, and severe stress.

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3. Hypothalamic-Pituitary-Gonadal (HPG) Axis

3.1 GnRH Pulsatility

GnRH is released in pulses (∼1 pulse per hour in humans):

$$\text{GnRH}(t) = A \sum_i \exp\left(-\frac{(t-t_i)^2}{2\sigma_{\text{pulse}}^2}\right) + \text{baseline} \tag{6}$$

Symbol	Meaning	Typical Value
A	Amplitude of GnRH pulse	0.5–10 nM
t_i	Time of i-th pulse	∼60 min apart
σ_{pulse}	Duration of pulse (Gaussian SD)	∼10–15 min
baseline	Interpulse baseline GnRH	∼0.01–0.1 nM

Explanation: GnRH pulsatility is essential for FSH/LH secretion. Too-slow pulses (<1/h) favor FSH; too-fast pulses (>2/h) favor LH. Pulse frequency is modulated by estradiol, progesterone, and kisspeptin neurons.

3.2 Estradiol-Driven LH Surge

During the follicular phase, rising estradiol triggers a massive LH surge (positive feedback):

$$[LH]_{\text{surge}} = [LH]_{\text{basal}} + A_{LH} \cdot f_{\text{surge}}(E_2) \tag{7}$$

where

$$f_{\text{surge}}(E_2) = \begin{cases} 0 & \text{if } [E_2] < E_2^{\text{thresh}} \\ \frac{([E_2]-E_2^{\text{thresh}})^n}{K_d^n + ([E_2]-E_2^{\text{thresh}})^n} & \text{if } [E_2] \geq E_2^{\text{thresh}} \end{cases} \tag{8}$$

Symbol	Meaning	Typical Value
$[LH]_{\text{basal}}$	Basal LH during follicular phase	2–5 mIU/mL
$[LH]_{\text{surge}}$	Peak LH during surge	30–100 mIU/mL
$[E_2]$	Plasma estradiol	100 (early) → 300–400 (surge)
E_2^{thresh}	Estradiol threshold for LH surge	∼200–250 pg/mL
K_d	Half-saturation for surge activation	∼50 pg/mL (above threshold)
n	Hill coefficient	2.0–3.0 (steep threshold)

Explanation: The LH surge (occurring ∼36 h before ovulation) is triggered by sustained estradiol elevation above ∼200 pg/mL for ∼48 h. The high cooperativity ($n \approx 2$) ensures a sharp, switch-like surge.

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4. Hypothalamic-Pituitary-Thyroid (HPT) Axis

4.1 TSH Suppression by T3/T4

TSH secretion is suppressed by circulating thyroid hormones:

$$\text{TSH} = \text{TSH}_{\max} \cdot \frac{1}{1 + \left(\frac{[T_3]+0.1[T_4]}{IC_{50}^{TSH}}\right)^{n_{TSH}}} \tag{9}$$

Symbol	Meaning	Typical Value
$[T_3]$	Plasma T3 concentration	4–8 pmol/L
$[T_4]$	Plasma T4 concentration	70–140 pmol/L
IC_{50}^{TSH}	T3/T4 concentration suppressing TSH by 50%	∼4 pmol/L
n_{TSH}	Hill coefficient	2.0–3.0

Symbol	Meaning	Typical Value
TSH (normal)	Normal TSH range	0.5–2.5 mIU/L

Explanation: Thyroid hormones exert negative feedback at the pituitary (direct suppression of TSH-secreting thyrotrophs) and hypothalamus (suppression of TRH). The factor 0.1 reflects that T4 is ∼10× less potent than T3. In primary hypothyroidism, TSH rises to 5–50+ mIU/L; in hyperthyroidism, TSH falls to <0.01 mIU/L.

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5. Neuroinflammation and Cytokine Signaling

5.1 Microglial Activation Dynamics

Activated microglia produce pro-inflammatory cytokines in a concentration-dependent manner:

$$\frac{d[Cytokine]}{dt} = k_{\text{prod}} \cdot [Stimulus]^n - k_{\text{deg}} \cdot [Cytokine] \tag{10}$$

At steady state:

$$[Cytokine]_{ss} = \frac{k_{\text{prod}}}{k_{\text{deg}}} \cdot [Stimulus]^n \tag{11}$$

Symbol	Meaning	Typical Value
k_{prod}	Cytokine production rate constant	0.1–1.0 (nmol/mL·min)
k_{deg}	Degradation rate constant	0.03–0.1 min^{-1}
$[Stimulus]$	Microglial activator concentration	Concentration-dependent
n	Hill coefficient	1.5–2.5

Explanation: Acute immune challenge (e.g., LPS from gram-negative bacteria) rapidly induces microglial TNF-α/IL-1β production (within minutes), peaking ∼2–4 h post-challenge. Chronic stimulation maintains elevated baseline cytokine levels, driving neuronal toxicity.

5.2 Cytokine-Mediated Neuronal Excitotoxicity

The relationship between cytokine concentration and NMDA calcium current:

$$I_{\text{Ca}}^{\text{NMDA}} = I_{\max} \cdot \frac{[Cytokine]^n}{EC_{50}^n + [Cytokine]^n} \tag{12}$$

Symbol	Meaning	Typical Value
$I_{\text{Ca}}^{\text{NMDA}}$	NMDA-mediated Ca^{2+} current	pA
$I_{\max}$	Maximum calcium current	100–300 pA
$[Cytokine]$	TNF-α or IL-1β concentration	nM
EC_{50}	Cytokine concentration for half-maximal effect	1–10 nM
n	Hill coefficient	1.5–2.0

Explanation: TNF-α and IL-1β increase neuronal NMDA receptor surface expression and enhance glutamate-induced Ca^{2+} influx, leading to excitotoxic cascade. This contributes to neuronal death in neuroinflammatory conditions.

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6. Gut-Brain Axis and Microbiota

6.1 Short-Chain Fatty Acid (SCFA) Production

Bacterial fermentation of dietary fiber produces SCFAs (butyrate, propionate, acetate):

$$v_{\text{SCFA}} = \frac{V_{\max}^{\text{SCFA}} \times [Fiber]}{K_m^{\text{Fiber}} + [Fiber]} \tag{13}$$

Symbol	Meaning	Typical Value
v_{SCFA}	SCFA production rate	mmol/day
$V_{\max}^{\text{SCFA}}$	Maximum SCFA production rate	250–350 mmol/day
$[Fiber]$	Effective fiber concentration in colon	1–2 mM
K_m^{Fiber}	Semisaturation constant for fiber	1–2 mM
Butyrate	Fraction of total SCFA	60% (acetate 20%, propionate 20%)
High-fiber diet	40 g/day fermentable fiber	∼90 mmol/day butyrate

Symbol	Meaning	Typical Value
Low-fiber diet	10 g/day fermentable fiber	∼20 mmol/day butyrate

Explanation: Dietary fiber (resistant starch, inulin, other non-digestible carbohydrates) is fermented by colonic bacteria, producing SCFAs. Butyrate is the major energy source for colonocytes and crosses the BBB via MCT1 transporter to exert neuroprotective effects (HDAC inhibition, anti-inflammatory signaling). Low butyrate production (dysbiosis, low-fiber diet) contributes to intestinal barrier dysfunction and reduced BDNF in hippocampus.

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7. Circadian Rhythm Dynamics

7.1 Molecular Clock (CLOCK/BMAL1 Transcription-Translation Feedback)

The core circadian clock operates through transcriptional and translational feedback:

$$\frac{d[PER]}{dt} = k_{\text{Per syn}} \cdot [CLOCK/BMAL1] - k_{\text{Per deg}} \cdot [PER] \tag{14}$$

$$\frac{d[CLOCK/BMAL1]}{dt} = k_{\text{CB syn}} - k_{\text{CB deg}} \cdot [CLOCK/BMAL1] - k_{\text{inhibit}} \cdot [PER] \cdot [CLOCK/BMAL1] \tag{15}$$

Symbol	Meaning	Typical Value
$[PER]$	Period protein concentration	Oscillating, 0–1 (normalized)
$[CLOCK/BMAL1]$	CLOCK/BMAL1 dimer concentration	Oscillating, 0–1 (normalized)
$k_{\text{Per syn}}$	Per gene synthesis rate	0.1 h^{-1}
$k_{\text{Per deg}}$	Per protein degradation rate	0.05 h^{-1}
$k_{\text{CB syn}}$	CLOCK/BMAL1 synthesis rate (basal)	0.05 h^{-1}
$k_{\text{CB deg}}$	CLOCK/BMAL1 degradation rate	0.03 h^{-1}

Symbol	Meaning	Typical Value
Period	Period of oscillation	24 h

Explanation: The molecular clock oscillates with a 24 h period due to negative feedback: CLOCK/BMAL1 activates Per and Cry gene transcription; PER/CRY proteins accumulate and inhibit CLOCK/BMAL1 activity; subsequent degradation of PER/CRY relieves inhibition, restarting the cycle. This generates a self-sustained oscillation despite no external cues.

7.2 Melatonin Synthesis and Degradation

Melatonin is synthesized from tryptophan in the pineal gland during darkness:

$$\frac{d[\text{Melatonin}]}{dt} = k_{\text{syn}}(t) \cdot [\text{Serotonin}] - k_{\text{deg}} \cdot [\text{Melatonin}] \tag{16}$$

where $k_{\text{syn}}(t)$ follows a circadian profile (high at night, low during day):

$$k_{\text{syn}}(t) = k_{\text{syn,max}} \cdot \left(1 - \cos\left(\frac{2\pi t}{24}\right)\right)/2 \tag{17}$$

Symbol	Meaning	Typical Value
[Melatonin]	Plasma melatonin concentration	10–50 pg/mL (night); <5 pg/mL (day)
$k_{\text{syn,max}}$	Maximum melatonin synthesis rate	0.5 h^{-1}
k_{deg}	Melatonin degradation rate	0.15 h^{-1}Serotonin (half-life 4.5 h)
	Pineal serotonin concentration	Serotonin> 1 μM

Explanation: Melatonin is synthesized exclusively at night (tryptophan → serotonin → N-acetylserotonin → melatonin by AANAT and ASMT enzymes). Light suppresses AANAT activity via a -adrenergic pathway (noradrenaline from sympathetic terminals). Melatonin rises ∼9 PM, peaks 2 AM, and falls by 7 AM, synchronizing peripheral oscillators and promoting sleep onset via MT1/MT2 receptors.

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8. Case Studies with Numerical Solutions

8.1 Case Study 1: BBB Glucose Transport During Hypoglycemia

Scenario: Model glucose transport across the BBB in fasting (normal glucose) vs. hypoglycemia using Michaelis-Menten kinetics.

Given:

- Normal blood glucose: $[\mathrm{S}]_{\mathrm{normal}} = 5$ mmol/L
- Hypoglycemic glucose: $[\mathrm{S}]_{\mathrm{hypo}} = 2$ mmol/L
- GLUT1 parameters: $V_{\max} = 150$ nmol·cm^{-2}·min^{-1}, $K_m = 1.7$ mmol/L

Solution:

Normal glucose (5 mmol/L):

$$v_{\mathrm{normal}} = \frac{150 \times 5}{1.7 + 5} = \frac{750}{6.7} \approx 112 \text{ nmol} \cdot \text{cm}^{-2} \cdot \text{min}^{-1}$$

Saturation: $\frac{[S]}{K_m+[S]} = \frac{5}{6.7} \approx 0.75$ (75% saturation — operating near saturation)

Hypoglycemic glucose (2 mmol/L):

$$v_{\mathrm{hypo}} = \frac{150 \times 2}{1.7 + 2} = \frac{300}{3.7} \approx 81 \text{ nmol} \cdot \text{cm}^{-2} \cdot \text{min}^{-1}$$

Saturation: $\frac{2}{3.7} \approx 0.54$ (54% saturation)

Transport reduction: $\frac{v_{\mathrm{hypo}}}{v_{\mathrm{normal}}} = \frac{81}{112} \approx 0.72$ (28% reduction in glucose transport)

Clinical implication: Even in moderate hypoglycemia (2 mmol/L), glucose transport across the BBB is reduced by 28%. Severe hypoglycemia (<1 mmol/L) would reduce transport further, causing altered cognition, seizures, loss of consciousness. GLUT1 deficiency syndrome causes recurrent hypoglycemia-like episodes despite normal blood glucose, demonstrating the critical importance of this transporter.

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8.2 Case Study 2: HPA Axis Activation and Recovery from Acute Stress

Scenario: Model cortisol dynamics during acute psychological stress (30 min) and recovery over 3 hours using cortisol production/clearance kinetics.

Model: Cortisol follows first-order kinetics with stress-driven enhancement:

$$\frac{d[Cortisol]}{dt} = k_{\mathrm{prod}}(t) - k_{\mathrm{clear}} \cdot [Cortisol](t)$$

where $k_{\mathrm{prod}}(t)$ includes both basal and stress-driven components.

Solution:

Basal state (pre-stress, 2 PM):

$$[Cortisol]_{\text{basal}} = 0.25\ \mu\text{mol/L}$$

$$\text{At steady-state}: k_{\text{prod}} = k_{\text{clear}} \times [Cortisol] = 0.02\ \text{min}^{-1} \times 0.25 = 0.005\ \mu\text{mol/L} \cdot \text{min}^{-1}$$

During stress (0–30 min): Stress multiplier = 5–10×

$$k_{\text{prod}}^{\text{stress}} = 5 \times 0.005 = 0.025\ \mu\text{mol/L} \cdot \text{min}^{-1}$$

$$[Cortisol]_{\text{stress}} = \frac{k_{\text{prod}}}{k_{\text{clear}}} = \frac{0.025}{0.02} = 1.25\ \mu\text{mol/L}$$

Time to reach peak (assuming exponential approach):

$$[Cortisol](t) = [Cortisol]_{\text{baseline}} + ([Cortisol]_{\text{peak}} - [Cortisol]_{\text{baseline}}) \times (1 - e^{-kt})$$

At $t = 30$ min ($k = 0.02\ \text{min}^{-1}$):

$$[Cortisol](30) = 0.25 + (1.25 - 0.25) \times (1 - e^{-0.02 \times 30}) = 0.25 + 1.0 \times (1 - 0.55) = 0.25 + 0.45 = 0.70\ \mu\text{mol/L}$$

Recovery (after stress ends):

$$[Cortisol](t) = 0.70 \times e^{-0.02(t-30)} + 0.25 \times (1 - e^{-0.02(t-30)})$$

At $t = 90$ min (60 min post-stress):

$$[Cortisol](90) = 0.70 \times e^{-1.2} + 0.25 \times (1 - e^{-1.2}) = 0.70 \times 0.30 + 0.25 \times 0.70 = 0.21 + 0.18 = 0.39\ \mu\text{mol/L}$$

Summary: Cortisol rises from 0.25 → 0.70 μmol/L during stress, then recovers to baseline over 1.5 hours. Prolonged stress without recovery impairs negative feedback (elevated baseline cortisol, flattened diurnal rhythm), contributing to HPA axis dysregulation seen in PTSD and depression.

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